Daniil Serebrennikov

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Education

Johns Hopkins University — Baltimore, USA
Ph.D. in Mathematics, Advisor: Vyacheslav Shokurov — 2022–2027 (Expected)

Johns Hopkins University — Baltimore, USA
M.A. in Mathematics — 2022–2023

Moscow Institute of Physics and Technology — Moscow, Russia
B.A. in Mathematics and Physics, with Honors — 2018–2022

Research interests

Higher-dimensional birational geometry, applications of Hodge Theory, boundedness and finiteness-type questions for Calabi–Yau varieties, moduli problems, and derived algebraic geometry.

Publications and Preprints

[1] D. Serebrennikov, *Constructible properties of minimal models*, 2026. In preparation.

[2] D. Serebrennikov, *Towards the global Torelli theorem*, 2026. arXiv:2608.27441.

[3] D. Serebrennikov, *Constructibility aspects of the cone conjecture*, 2026. arXiv:2604.27303.

[4] D. Serebrennikov, *On the finiteness of log surfaces*, Int. J. Math. **37** (2026), no. 13, 2650065.

Awards

- Professor Joel Dean Award for Excellence in Teaching — 2025
- Enhanced State Academic Scholarship for Leadership and Service — 2021–2022
- Abramov–Frolov Foundation Scholarship for Academic Excellence — 2018–2021

Invited Talks

Constructibility aspects of the cone conjecture

- **The Pennsylvania State University** — Nov 2026
  Algebra and Number Theory Seminar
- **Rutgers University** — Oct 2026
  Algebra Seminar
- **Purdue University** — Sep 2026
  Algebraic Geometry Seminar
- **Duke University** — Sep 2026
  Algebraic Geometry Seminar

On the finiteness of log surfaces

- **University of Illinois Chicago** — Jan 2026
  Algebraic Geometry Seminar

Contributed presentations

Short Talks

- **Rutgers University** May 2026
  AGNES

Poster Sessions

- **Brown University** Nov 2026
  AGNES
- **ICERM** Nov 2026
  Moduli of K3 surfaces
- **Simons Foundation** Oct 2026
  Simons Annual Meeting
- **University of Massachusetts Amherst** Nov 2025
  AGNES

Expository Talks

University of Illinois Chicago

- **Graduate Algebraic Geometry Seminar** Apr 2026
  Towards a moduli space of Calabi–Yau varieties
- **Toric Compactifications Learning Seminar** Apr 2026
  The b-semiampleness conjecture
- **Birational Geometry Learning Seminar** Apr 2026
  The Non-vanishing and Basepoint-free Theorems

Northwestern University

- **Informal Algebraic Geometry Oberseminar** Feb 2026
  The Cone Conjecture, Calabi–Yau varieties and All That

Purdue University

- **Pre-talk for Algebraic Geometry Seminar** Sep 2026
  Kodaira dimension

Johns Hopkins University

- **Algebraic Geometry Seminar** Sep 2026
  Finiteness, Boundedness, and Constructibility for Calabi–Yau varieties
- **Algebraic Geometry Preprint Seminar** Apr 2026
  On L-equivalence for K3 surfaces and hyperkähler manifolds, preprint
- **Algebraic Geometry Preprint Seminar** Oct 2025
  Baily–Borel compactifications of period images and the b-semiampleness conjecture, preprint
- **Graduate Student Seminar** Sep 2025
  What is a K3 surface?
- **Graduate Student Seminar** Apr 2025
  Constructibility in Algebraic Geometry
- **Algebraic Geometry Preprint Seminar** Mar 2025
  Boundedness of elliptic Calabi–Yau threefolds, preprint
- **Graduate Student Seminar** Feb 2025
  Rigidity in Algebraic Geometry

- **Algebraic Geometry Preprint Seminar** Mar 2024
  Moduli of boundary polarized Calabi–Yau pairs, preprint
- **Graduate Student Seminar** Feb 2024
  Singularities of varieties
- **Graduate Student Seminar** Dec 2023
  Birational Geometry and Calabi–Yau varieties
- **Topics in Algebraic Geometry** Dec 2023
  Boundedness of log Calabi–Yau pairs of Fano type, preprint
- **Algebraic Geometry Seminar** Nov 2022
  Finiteness of log Calabi–Yau surfaces
- **Goodwillie Calculus Learning Seminar** Oct 2022
  Cubic Diagrams
- **What is... Seminar** Sep 2022
  What is birational geometry?

Moscow Institute of Physics and Technology

- **Summer Mini-Course** Jul 2024
  Introduction to Birational Geometry, 4 lectures
- **Pro-Seminar** Jun 2023
  A cone conjecture for log Calabi–Yau surfaces, preprint

Conferences/Workshops Attended

- **Algebraic Geometry Northeastern Series (AGNES)** Nov 2026 (upcoming)
  Brown University
- **Algebra and Number Theory Day** Nov 2026 (upcoming)
  University of Maryland
- **Workshop: Moduli of K3 surfaces** Nov 2026 (upcoming)
  ICERM
- **Recent Outcomes in Birational and Complex Algebraic Geometry** Oct 2026
  Stony Brook University and the Simons Center for Geometry and Physics
- **Simons Annual Meeting: Moduli of Varieties** Oct 2026
  Simons Foundation
- **Forty Years of Higher Dimensional Geometry** Jun 2026
  Princeton University
- **Foliations and Birational Geometry** May 2026
  University of Michigan
- **Algebraic Geometry Northeastern Series (AGNES)** May 2026
  Rutgers University
- **Algebra and Number Theory Day** Apr 2026
  Johns Hopkins University
- **Algebraic Geometry Northeastern Series (AGNES)** Nov 2025
  University of Massachusetts Amherst
- **Algebra and Number Theory Day** Nov 2025
  University of Maryland
- **Simons Annual Meeting: Moduli of Varieties** Oct 2025
  Simons Foundation
- **I. R. Shafarevich Memorial Lectures** Jun 2025

Steklov Mathematical Institute

- **Georgia Algebraic Geometry Symposium** Mar 2025
  University of Georgia
- **Winter school on enumerative geometry and mirror symmetry** Jan 2025
  California Institute of Technology
- **K-trivial Varieties and Their Moduli** Jan 2025
  University of California San Diego
- **Algebra and Number Theory Day** Nov 2024
  Johns Hopkins University
- **Summer school on classical, derived and condensed geometry** July 2024
  Moscow Institute of Physics and Technology
- **Algebraic Geometry Northeastern Series (AGNES)** Mar 2024
  Boston College
- **Winter School on New Applications of Mixed Hodge Modules** Jan 2024
  Simons Center for Geometry and Physics
- **Higher Dimensional Algebraic Geometry: James 60** Jan 2024
  University of California San Diego
- **I. R. Shafarevich Centenary Conference** Jun 2023
  Steklov Mathematical Institute
- **Algebra and Number Theory Day** Apr 2023
  Johns Hopkins University
- **Algebra and Number Theory Day** Nov 2022
  University of Maryland

Teaching

Johns Hopkins University

- **Differential Equations** Spring 2024, Fall 2024, Spring 2025 (Head TA)
  Teaching Assistant
- **Linear Algebra** Fall 2025 (Head TA)
  Teaching Assistant
- **Calculus III** Fall 2023
  Teaching Assistant
- **Algebra I, II** Spring 2024, Fall 2026
  Grader
- **Introduction to Algebraic Geometry** Fall 2026
  Grader

Moscow Public School No. 1560

- **Algebra and Geometry** 2021–2022
  High School Math Teacher

Professional Service

- Reviewer for zbMATH Open
- Mentor for Directed Reading Program

Leadership & Service

Student Council, Department of General and Applied Physics (DGAP), Moscow Institute of Physics and Technology (MIPT)

- Chair 2021–2022
- Vice Chair, Lead Student Housing Coordinator 2020–2021
- Head of the Student Housing Committee, Senior Student Housing Coordinator 2019–2020
- Co-organizer of MIPT Open Day, Phystech Olympiad, Commencement Ceremony, etc. 2018–2019

Miscellaneous

Winner of Russian Physics/Mathematics Olympiads for High School students (2017–2018).

Far Eastern Federal University Vladivostok, Russia
Laboratory of Thin Film Technologies, Advisor: Aleksandr Davydenko 2015–2017

- Measured the surface morphology and magnetic domain structure of a Pd/Co/Pd thin film with perpendicular magnetic anisotropy.
- Established an empirical relationship between perpendicular magnetic anisotropy and the thicknesses of the lower Pd and Co layers in the polycrystalline Pd/Co/Pd thin film.
- Developed LabVIEW software to automate data acquisition and thickness monitoring for a quartz crystal microbalance in the Laboratory of Thin Film Technologies.

Languages: Russian (native), English (fluent), French (elementary), Mathematics (proficient).